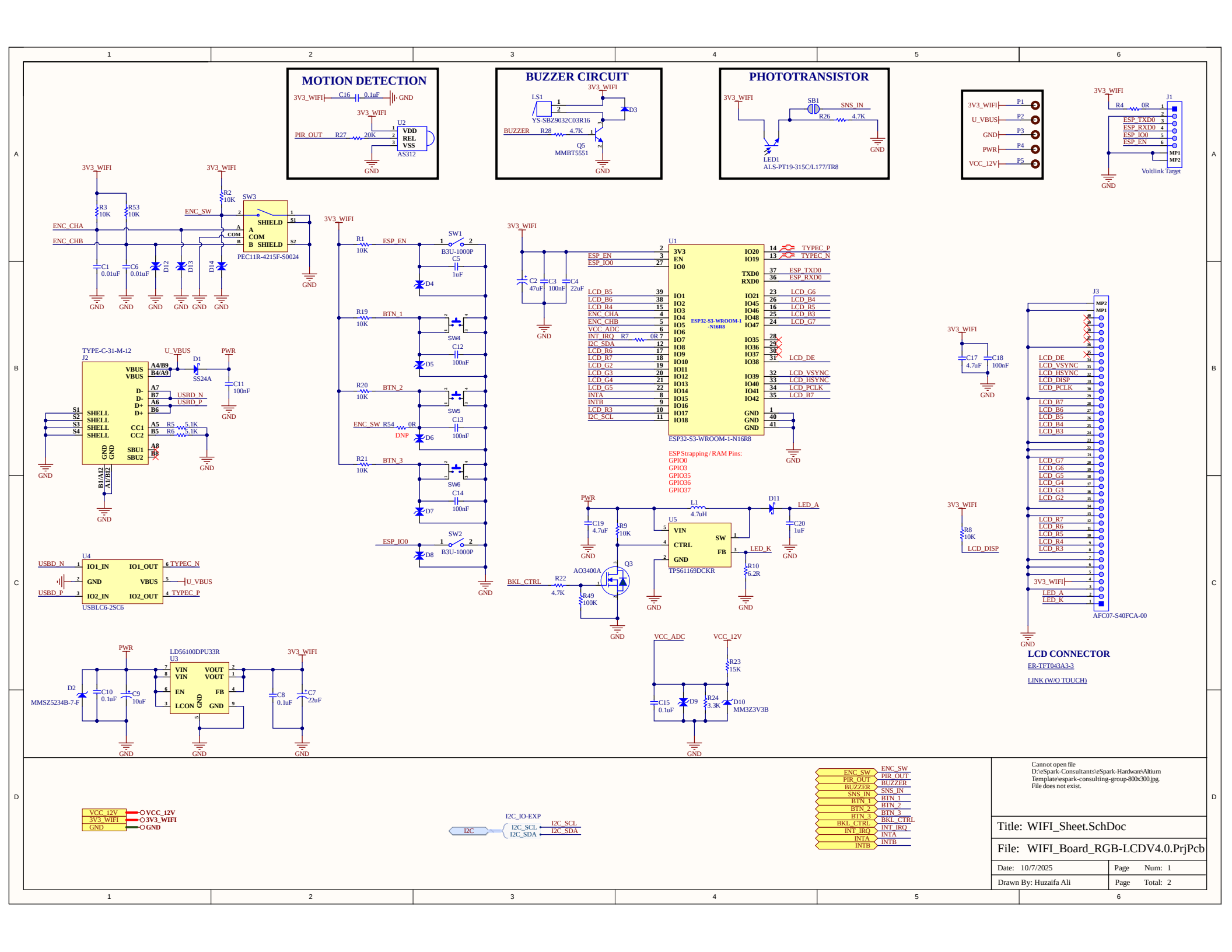
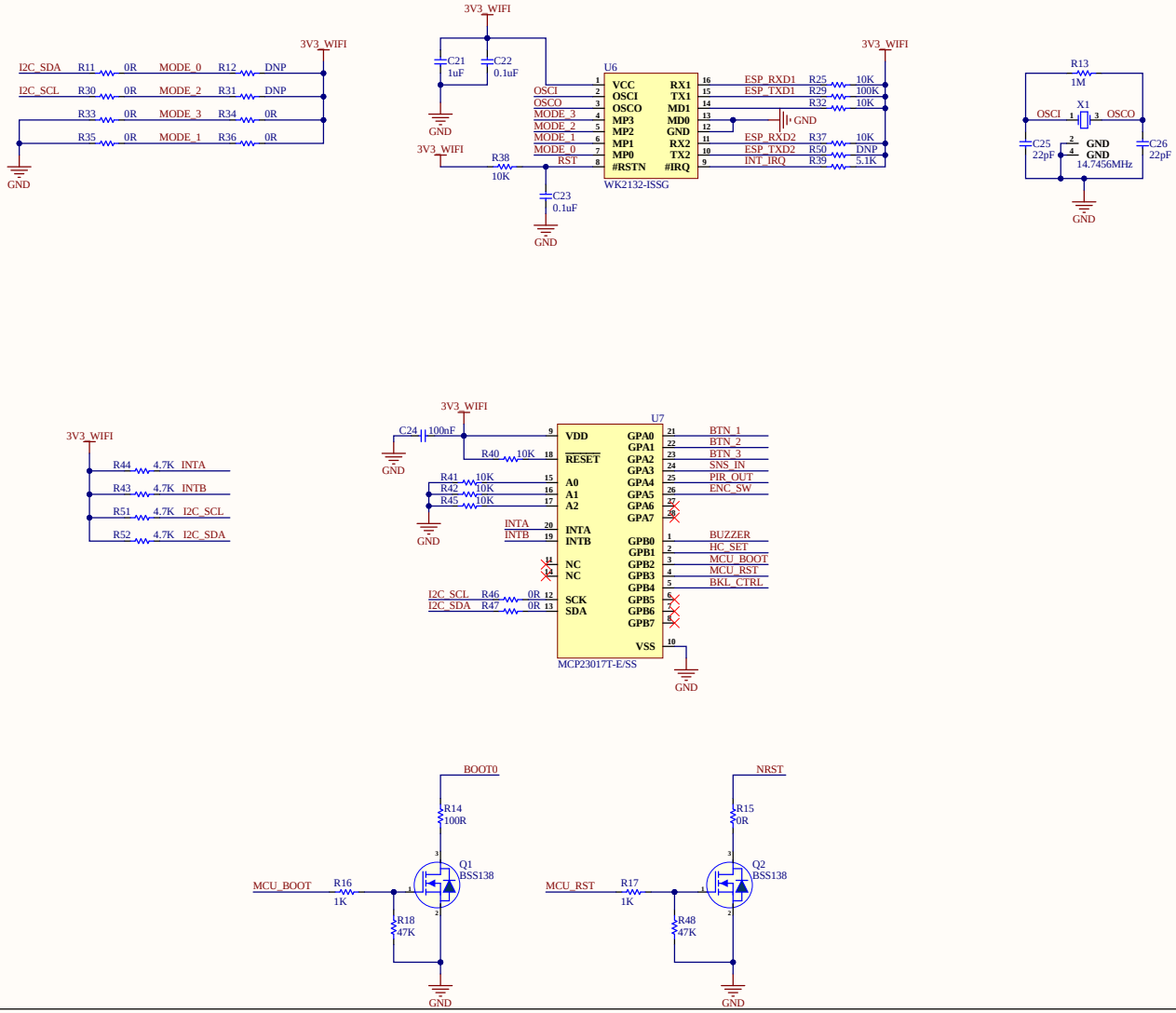
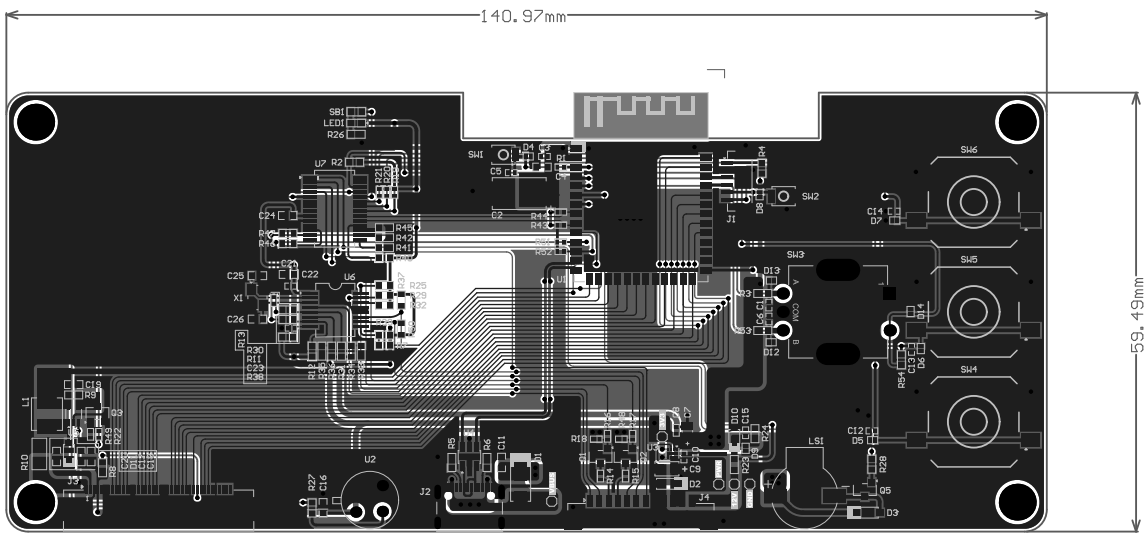

MOTION DETECTION
BUZZER CIRCUIT
PHOTOTRANSISTOR
ESP32-S3-WROOM-1-N16R8
ESP Strapping / RAM Pins:
GPIO0
GPIO3
GPIO35
GPIO36
GPIO37
LCD CONNECTOR
ER-TFT043A3-3
LINK (W/O TOUCH)
Cannot open file D:\eSpark-Consultants\eSpark-Hardware\Altium Template\espark-consulting-group-800x300.jpg. File does not exist.
Title: WIFI_Sheet.SchDoc
File: WIFI_Board_RGB-LCDV4.0.PrjPcb
Date: 10/7/2025
Page Num: 1
Drawn By: Huzaifa Ali
Page Total: 2



Cannot open file D:\eSpark-Consultants\eSpark-Hardware\Altium Template\espark-consulting-group-800x300.jpg. File does not exist.

Title: IO_Expander_Sheet.SchDoc	
File: WIFI_Board_RGB-LCDV4.0.PrjPcb	
Date: 10/7/2025	Page Num: 2
Drawn By: Huzaifa Ali	Page Total: 2



Layer	Name	Material	Thickness	Constant	Board Layer Stack
	Top Overlay				
	Top Solder	Solder Resist	0.010mm	3.5	
1	Top Layer		0.035mm		
	Dielectric 1	FR-4	1.500mm	4.6	
2	Bottom Layer		0.035mm		
	Bottom Solder	Solder Resist	0.010mm	3.5	
	Bottom Overlay				
Total board thickness:			1.590mm		

Comment	Description	Designator	Footprint	LibRef	Quantity
0.01uF	CAP CER 10000PF 25V X5R 0402	C1, C6	Cap-0402	CC0402KRX5R8BB103	2
47uF	CAP TANT 47UF 10% 25V 2917	C2	Case-D	T495D476K025ATE250	1
100nF	CAP CER 0.1UF 25V X5R 0402	C3, C12, C13, C14	Cap-0402	CC0402KRX5R8BB104	4
22uF	CAP CER 22UF 10V X5R 0603	C4	Cap-0603	CC0603MRX5R6BB226	1
1uF	CAP CER 1UF 16V X7S 0402	C5	Cap-0402	GRM155C71C105KE11 D	1
22uF	22uF 10V 5G ±20% 0805 Tantalum Capacitors ROHS	C7	Tant-0805	TCTP1A226M8R	1
0.1uF	CAP CER 0.1UF 16V X7R 0402, CAP CER 0.1UF 50V X5R 0402	C8, C10, C15, C22, C25	Cap-0402	CC0402RKX7R7BB104, C1005X5R1H104K050B B	5
10uF	CAP TANT 10UF 10% 16V 1206	C9	Case-A	T491A106K016AT	1
100nF	CAP CER 0.1UF 50V X5R 0603	C11, C18, C24	Cap-0603	CC0603KRX5R9BB104	3
0.1uF	CAP CER 0.1UF 50V Y5V 0603	C16	Cap-0603	CC0603ZRY5V9BB104	1
4.7uF	CAP CER 4.7UF 16V X5R 0603	C17, C19	Cap-0603	CC0603KRX5R7BB475	2
1uF	CAP CER 1UF 50V X5R 0603, 1uF 25V X5R ±20% 0603 Multilayer Ceramic Capacitors MLCC - SMD/SMT ROHS	C20, C21	Cap-0603	CL10A105KB8NNNC, TCC0603X5R105M25D CT	2
22pF	CAP CER 22PF 50V C0G/NP0 0603	C25, C26	Cap-0603	CL10C220JB8NNNC	2
SS24A	40V 500mV@2A 2A SMA(DO-214AC) Schottky Barrier Diodes (SBD) ROHS	D1	SMA	SS24A	1
MMSZ5234B-7-F	DIODE ZENER 6.2V 500MW SOD123	D2	SOD-123	MMSZ5234B-7-F	1
1N4148W-7-F	DIODE GEN PURP 100V 300MA SOD123	D3	SOD-123	1N4148W-7-F	1
LESD8D3.3CAT5G	5V 10V 3.3V Bi-Directional SOD-882 ESD Protection Devices ROHS	D4, D5, D6, D7, D8, D9, D12, D13, D14	SOD-882 Bi Directional	LESD8D3.3CAT5G	9
MMSZ3V3B	DIODE ZENER 3.3V 200MW SOD323F	D10	SOD323F	MMSZ3V3B	1
NSR0240HT1G	DIODE SCHOTTKY 40V 250MA SOD323	D11	SOD-323	NSR0240HT1G	1
Volttlink Target	1x9P 3H 1 1mm 6 SMD,P=1mm Wire To Board / Wire To Wire Connector ROHS	J1	Volttlink Target	C145963	1
TYPE-C-31-M-12	5A 1.98P Female Type C SMD USB Connectors ROHS	J2	TYPE-C-31-M-12_	TYPE-C-31-M-12	1
AFC07-S40FCA-00	Slide Lock 40P Bottom Contact Surface Mount, Right Angle 0.5mm SMD,P=0.5mm,Surface Mount,Right Angle FFC, FPC (Flat Flexible) Connector Assemblies ROHS	J3	AFC07-S40FCA-00	AFC07-S40FCA-00	1
HC-1.25-12PLT	1x12P 12P PicoBlade(MX 1.25) Tin 12 -25°C~+85°C 2A 1 1.25mm Brass Surface Mount,Vertical SMD,P=1.25mm Wire To Board Connector ROHS	J4	HC-1.25-12PLT	HC-1.25-12PLT	1
4.7uH	Fixed Ind 4.7UH 2.8A 105MOHM SMD	L1	MWSA0402S-4R7MT	MWSA0402S-4R7MT	1
ALS-PT19-315C/L177/TR8	SENSOR PHOTO 630NM 20MO	LED1	ALS-PT19-315C/L177/TR	ALS-PT19-315C/L177/TR8	1
YS-SBZ9032C03R16	Externally Driven 85dB@3V, 10cm 3V 2700Hz SMD,D=9mm Buzzers ROHS	LS1	YS-SBZ9032C03R16_	YS-SBZ9032C03R16	1
TEST POINT 0.5mm Dia	Test Point ON	P1, P2, P3, P4, P5	Test-point - 0.5mm	TEST POINT 0.5mm Dia	5
BSS138	50V 220mA 3.5Ω@10V, 220mA 225mW 1.6V@1mA N Channel SOT-23 MOSFETs ROHS	Q1, Q2	Sot-23	BSS138	2
AO3400A	30V 5.8A 32mΩ@10V, 5.8A 1.4W 1.4V@250uA N Channel SOT-23 MOSFETs ROHS	Q3	sot-23	AO3400A	1
MMBT5551	TRANS NPN 160V 0.6A SOT-23	Q5	SOT-23	MMBT5551	1
10K	RES 10K OHM 1% 1/10W 0402	R1, R19, R20, R21	Res-0402	RC0402FR-0710KL	4
10K	RES 10K OHM 1% 1/10W 0603, RES SMD 10K Ohm 1% 1/10W 0603	R2, R3, R8, R9, R25, R32, R37, R38, R40, R41, R42, R45, R53	Res-0603	RK73G1JTTD1002F, RC0603FR-0710KL	13
0R	RES 0 OHM JUMPER 1/10W 0603	R4, R7, R11, R30, R33, R34, R35, R36, R46, R47	Res-0603	RC0603JR-070RL	10
5.1K	RES 5.1K OHM 1% 1/10W 0603	R5, R6, R39	Res-0603	RC0603FR-075K1L	3
6.2R	250mW Thick Film Resistors 200V ±5% ±200ppm/°C 6.2Ω 1206 Chip Resistor - Surface Mount ROHS	R10	Res-1206	FRC1206J6R2 TS	1
DNP	RES 5.1K OHM 1% 1/10W 0603, RES 100K OHM 1% 1/10W 0603, RES 0 OHM JUMPER 1/10W 0603	R12, R31, R50, R54	Res-0603	RC0603FR-075K1L, RC0603FR-07100KL, RC0603JR-070RL	4
1M	RES 1M OHM 1% 1/16W 0402	R13	Res-0402	RC0402FR-071ML	1
100R	RES 100 OHM 1% 1/16W 0402	R14	Res-0402	RC0402FR-07100RL	1
0R	RES 0 OHM JUMPER 1/16W 0402	R15	Res-0402	RC0402FR-070RL	1
1K	RES 1K OHM 5% 1/16W 0402	R16, R17	Res-0402	RC0402JR-071KL	2
47K	RES 47K OHM 1% 1/16W 0402	R18, R48	Res-0402	RC0402FR-0747KL	2
4.7K	RES 4.7K OHM 1% 1/16W 0402	R22, R43, R44, R51, R52	Res-0402	RC0402FR-074K7L	5
15K	RES 15K OHM 1% 1/10W 0603	R23	Res-0603	RC0603FR-0715KL	1
3.3K	RES 3.3K OHM 0.5% 1/10W 0603	R24	Res-0603	RC0603DR-073K3L	1
4.7K	RES SMD 4.7K OHM 1% 1/10W 0603	R26, R28	Res-0603	RC0603FR-074K7L	2
20K	RES 20K OHM 1% 1/10W 0603	R27	Res-0603	RC0603FR-0720KL	1
100K	RES 100K OHM 1% 1/10W 0603	R29	Res-0603	RC0603FR-07100KL	1
100K	RES 100K OHM 1% 1/16W 0402	R49	Res-0402	RC0402FR-07100KL	1
B3U-1000P	SWITCH TACTILE SPST-NO 0.05A 12v	SW1, SW2	B3U-1000P	B3U-1000P	2
PEC11R-4215F-S0024	ENCODER MECH QUAD VERT PC PIN	SW3	PEC11R-4215F-S0024	PEC11R-4215F-S0024	1
TL3300CF160Q	SWITCH TACTILE SPST-NO 0.05A 12V	SW4, SW5, SW6	TL3300CF160Q_	TL3300CF160Q	3
ESP32-S3-WROOM-1-N16R8	SMD MODULE, ESP32-S3R8 WITH 3.3V	U1	41-SMD Module	ESP32-S3-WROOM-1-N16R8	1
AS312	Plugin Human Body Sensing Sensor ROHS	U2	AS312	AS312	1
LDS6100DPUS3R	IC REG LINEAR 3.3V 1A 8DFN	U3	DFN-8(1.2x1.6)	LDS6100DPUS3R	1
USBLC6-2SC6	6A 12V 150W 8V Unidirectional 5v SOT-23-6 ESD and Surge Protection (TVS/ESD) ROHS	U4	SOT-23-6	USBLC6-2SC6	1
TP5611691DCKR	IC LED DRV REGTR PWM 1.8A SC70-5	U5	SC70-5	TP5611691DCKR	1
WK2132-ISSG	SSOP-16-208mil Interface - Specialized ROHS	U6	SSOP-16-208mil	WK2132-ISSG	1
MCP23017T-E/SS	IC XPNDR 1.7MHZ I2C 28SSOP	U7	SSOP-28-208mil	MCP23017T-E/SS	1
14.7456MHz	14.7456MHz Surface Mount Crystal 20pF ±10ppm ±30ppm SMD3225-4P Crystals ROHS	X1	SMD3225-4P	SX3B14.7456F2010F30	1